

CS 315 - Cloud Computing (Midterms)

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Module 5: Virtualization

5.1 Introduction to Virtualization

_____. The act of providing access to resources and storage to multiple organizations. There is a problem when organizations requests resources for which the physical servers cannot handle.

_____. It refers to creating virtual copies of the server, network, operating system, or resources and sharing it among multiple servers. Each virtual copy can be distributed to organizations.

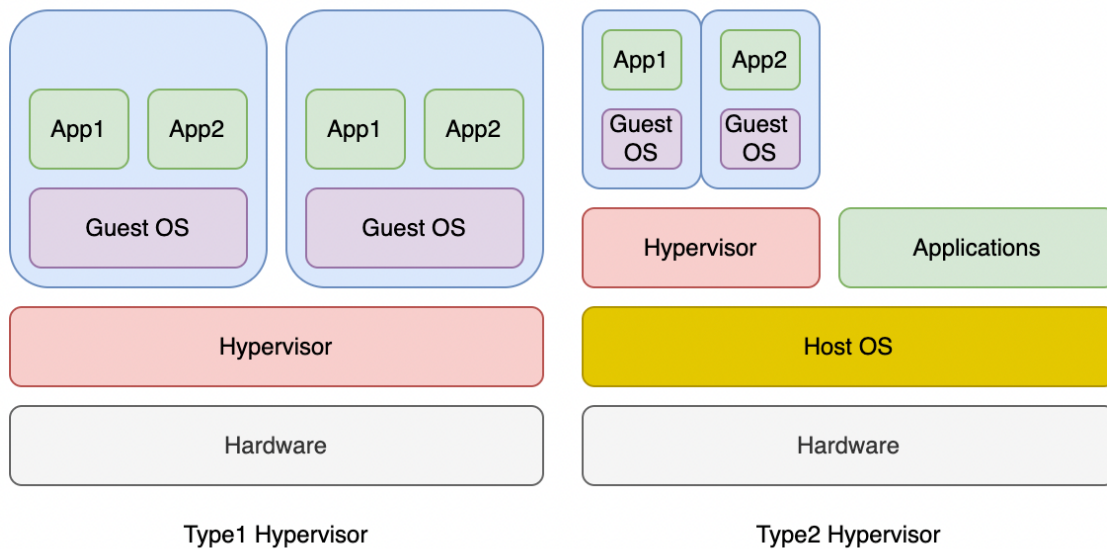


Figure 1. Virtualization of a physical resource

- _____. The machine which the virtual machine is created.
- _____. The virtual copy of a physical resource in a host machine.
- _____. The firmware that manages and controls virtual machines.

A single physical resource can be divided into multiple logical partitions and shared among organizations. Each physical resource is assigned a logical name for its address. When a user demands for a resource, a pointer is assigned to that physical resource.

Virtualization is helpful for enabling full utilization of infrastructure and overcome the problem of workload on the server or network.

As a result, virtualization is the foundation of cloud computing.

5.2 Features of Virtualization

Virtualization is considered to be the backbone of cloud computing because of the following features:

1. _____. Virtualization technology that divides the available resources into multiple partitions and provides access among multiple organizations.
2. _____. It keeps each virtual environment separate from each other to avoid conflicts. Damage from one virtual machine will not affect other virtual machines nor the host machine.
3. _____. The ability to scale up/down based on demand. For efficient growth of a business.

4. _____. Virtual technology allows organizations to be flexible which enables the availability of resources or applications.
5. _____. Virtualization provides a high level of security and protection to guest machines. It has firewalls and encryption for ensuring the security of data and applications. It protects the host machine from any harm or damage caused by the guest machines.

5.3 Benefits of Virtualization

The following are the benefits of virtualization:

1. _____. Virtualization gives us the ability of sharing physical resources. It provides us with as many resources we want anytime and anywhere. Hence, there is no need to buy those resources nor maintain them.
2. _____. Every time that a user demands for a resource or an application, the cloud vendor provides him with its virtual form without wasting time by installing and setting up the new resource.
3. _____. If there is a need for recovery, virtualization can create new resources instead of repairing old one with the help of backup facility.
4. _____. All operations are managed by third-party providers which simplifies the management of data center (e.g., storage, security, infrastructure).
5. _____. Hardware and software maintenance are handled by the cloud providers. Teams can focus on solving the company's problem.
6. _____. Virtualization enables the availability of a huge amount of resources quickly in comparison to physical resources. You can provision them quickly whenever needed as per the demands of the organizations.

5.4 Difference between Cloud Computing and Virtualization

Virtualization creates a virtual instance of a resource from a single hardware.

Cloud computing is focused on distributing those virtual instances to organizations based on demand.

Virtualization	Cloud Computing
Virtualization is a process of creating multiple copies of hardware and software on the same machine It is very simple to set up virtualization software.	Cloud computing is the on-demand access of hardware and software resources. Setting up a cloud environment is a complicated task.
Users need to have proper authentication before accessing virtual resources. Virtualization is more cost-efficient than cloud computing.	Any user can access cloud resources anywhere and anytime. Cloud computing can be more costly than virtualization.

5.5 Types of Virtualization

Virtualization can be divided into the following types:

1. _____. Creates virtual physical hardware resources so organizations can access the resources. It has three types:
 - **Full Virtualization.** This complete simulation allows a guest operating system (one being virtualized) to run in an isolated instance. The guest operating system is unmodified and each system call is directly executed on the hardware.

- **Para-Virtualization.** A modified and recompiled version of the guest operating system is made to run in the virtual machine. The guest operating system is modified and each system call is communicated on **Hypervisor** (or *Virtual Machine Manager (VMM)*).
 - **Hardware Assisted Virtualization.** Developers used to experience a lot of overhead on the system while creating VMM using software. Hence, they started using hardware for implementing the virtualization functions. So, under this type, the hypervisor sends the call to the processor. Then, it is the processor's job to create and maintain the virtual machines. This reduces the overhead on the system which enables the host system to host a large number of VMs. This increases the performance of VMs. Hardware assisted virtualization was introduced firstly by IBM in
2. _____. Creates virtual environments or applications on top of an existing host operating system. It has three types:
 - **Operating System Virtualization.** In operating system virtualization, multiple operating systems can run on a single set of hardware resources. Each OS is isolated.
 - **Application Virtualization.** Application virtualization includes hosting individual applications in a virtual environment which is separate from the native operating system. This technology encapsulates the computer program within the operating system.
 - **Service Virtualization.** Service virtualization refers to the simulation of those components that are unavailable during testing. The reason for their unavailability can be either that these resources are under development, blocked, restricted or owned a by third-party, or undergoing maintenance procedure.
 3. _____. Partition a single physical servers into multiple virtual servers to increase the utilization of a resource. Organizations access the resource in virtual form.
 4. _____. Many storage devices are pooled into a single storage then partitioned into logical volumes to be accessed by organizations.
 5. _____. Many versions of operating system runs on a single system simultaneously (*will be explored in chapter 5.7 Storage Virtualization*).
 6. _____. Creates virtual networks that allow the network administrator to automate many tasks. It is the process of combining hardware, software, and network functionality into a single virtual network. This virtual network is responsible for deploying and managing network resources. Network is very beneficial as it helps in improving productivity and increasing network speed, reliability, flexibility, and security.

5.6 Types of Hypervisor (or Virtual Machine Manager)

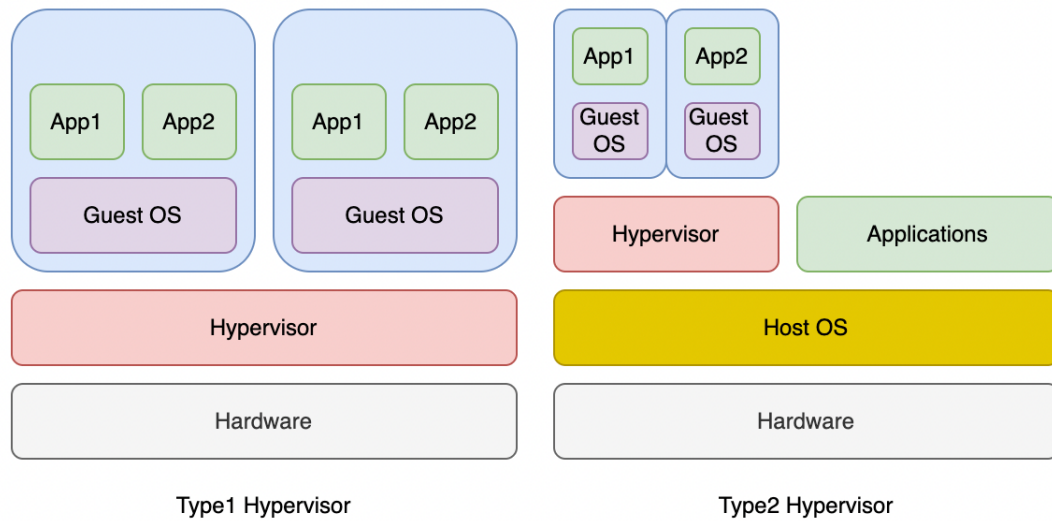


Figure 1. Virtualization of a physical resource

A firmware known as _____ control programs in virtualization. There are two types of Hypervisors:

1. _____ (or *bare-metal/native hypervisor*). Runs directly on the host hardware system. Its main job is to control the hardware resources and manage the guest operating system. It is the most basic form of hypervisor which enables efficient functioning and excellent performance. Examples include VMware vSphere, KVM, Oracle VM, and Citrix Hypervisor.
2. _____ (or *hosted hypervisor*). This type of hypervisor also functions as a management console for all VMs. Runs as a second layer, while the guest operating system runs as a third layer above the hardware. It runs on top of the host operating system. Examples include Oracle VM VirtualBox, VMware Workstation Pro, and Windows Virtual PC.

When you install Type-1 Hypervisor, it wipes out your entire computer operating system and turns your machine into a command-line interface meant to be managed remotely from another computer. When you install Type-2 Hypervisor, it runs on top of the operating system. It is meant for daily drive.

5.7 Storage Virtualization

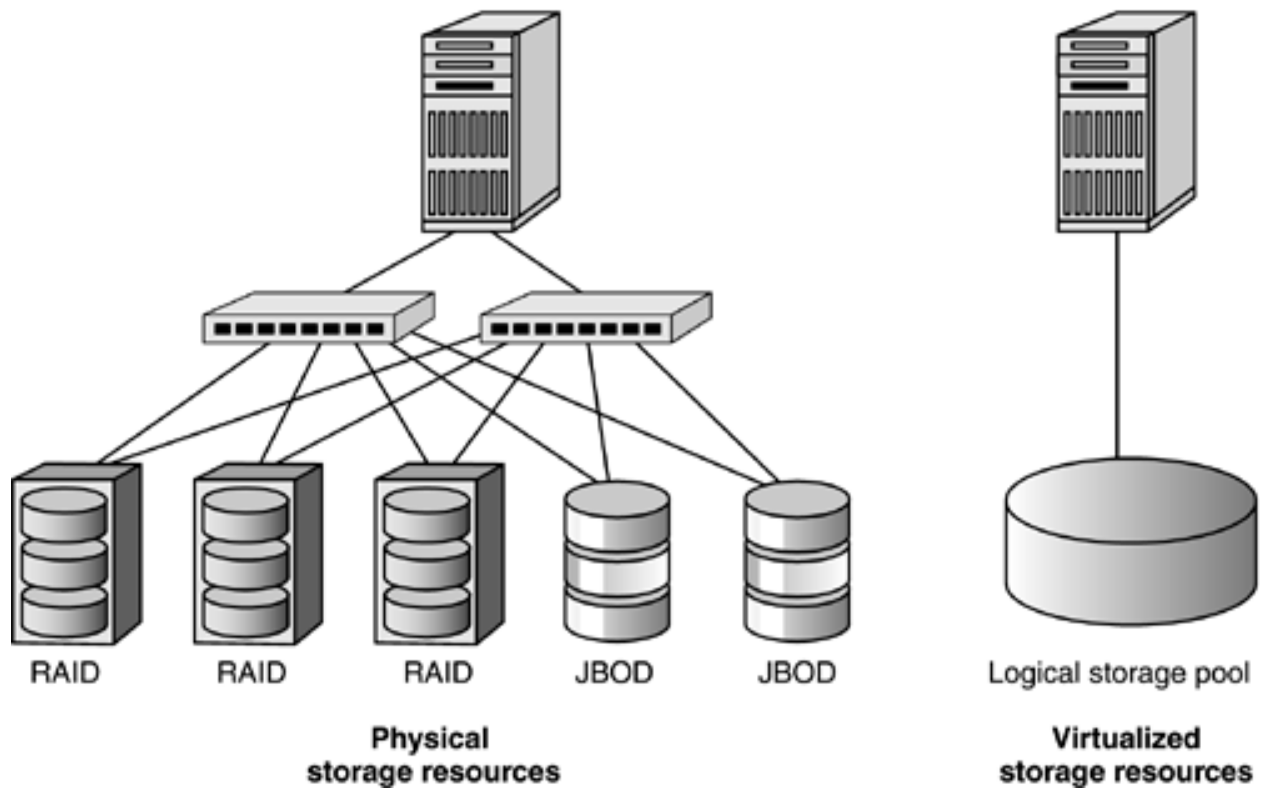


Figure 2. Comparison of physical storage resources and virtualized storage resources

Storage Virtualization (or *cloud storage*). Many storage devices are pooled into a single storage then partitioned into logical volumes to be accessed by organizations.

5.6.1 Characteristics of Storage Virtualization

The various characteristics that make storage virtualization so beneficial are as follows:

- Users are not concerned about where the data is stored and the speed of each physical hard disk drive, instead focuses on logical partition (virtualization of storage resources).
- It groups various storage devices from multiple vendors into a single storage pool and then allocates independent storage spaces to all the users.
- It provides automated storage optimization and management.
- There is centralized administration (central dashboard) for allocation of virtual storage spaces. There is no need to check each drive manually.

5.6.2 Types of Storage Virtualization

5.6.2.1 Block Virtualization

This is the most common type of storage virtualization.

Separates logical storage from physical storage. Users or hosts access data through a virtual path without interacting with the physical disks. When a user requests data, the system automatically uses a logical path to reassemble all the scattered data blocks and present them as a unified file.

Storage is divided into raw, individual blocks that act as independent hard drives. The server storage system manages these blocks and allocates multiple blocks to users as needed.

Commonly deployed in _____ and cloud storage. It provides fast, efficient, and reliable data storage suitable for almost any application (e.g., file storage, databases).

Data is accessed through **Fibre Channel**. Protocols like **iSCSI**, **Fibre Channel**, and **FCoE (Fibre Channel over Ethernet)** are used to make these storage blocks visible and accessible to server operating systems.

Each storage block can run a different operating system. A SAN can store these blocks across different physical systems running different OS platforms.

5.6.2.2 File Virtualization

Organizes and stores data as a centralized hierarchy of files and folders rather than raw blocks. Virtualization entities handle all operations to provide virtual storage for operating systems and applications.

Users, clients, and system administrators can easily access and retrieve data through a standard structure of directory trees, folders, and files. It allows bulk data access and provides a platform for very fast file recovery.

Commonly deployed in _____ systems and standard hard drives. It uses the same concepts as a network file system to provide a centralized, highly available, and easily accessible storage space.

Attached directly to the network. It operates using common file-level protocols like **NTFS** for Windows and **NFS** for Linux.

It is simpler to use, implement, and is less expensive compared to block-level storage. Because it is highly cost-effective, it is widely used by many organizations for applications that access data in the form of files rather than raw blocks.

5.6.3 Benefits of Storage Virtualization

The various benefits of storage virtualization are:

- _____. Allows the creation of storage pools that can easily expand or contract based on application demand. Users have complete flexibility to scale the number of physical storage devices up or down.
- _____. Data can be saved and retrieved as easily and effectively as working on a local computer. Users only require an internet connection and cloud access to manage their files.
- _____. Data transfers smoothly and quickly without traditional physical movement barriers. It ensures minimal to zero downtime when moving data to active applications.
- _____. Gives administrators the precise power to increase or decrease total storage capacity on demand based on software requirements.
- _____. Provides easy and secure management via robust security tools and protocols. Data is distributed across multiple locations, allowing applications to seamlessly bypass isolated system faults.
- _____. Maximizes utilization by automatically tiering data. It stores frequently accessed data on high-performance systems and routes rarely used data to slower, cost-effective systems.
- _____. Protects against data loss by maintaining more than one copy of the data across different virtual storage locations. If a storage device fails completely, users can instantly recover their data from another location.

Module 6: Data Storage

Storage Virtualization (or *cloud storage*). Many storage devices are pooled into a single storage then partitioned into logical volumes to be accessed by organizations. Examples of cloud storage include:

- Google Docs allows users to upload documents, worksheets, and presentation in Google data servers.
- Web e-mail providers such as Gmail, Hotmail, and Yahoo! Mails are stored in servers.
- Social networking sites allow us to store pictures and videos online.
- Sites like Pinterest and Youtube, allows users to upload and share videos with millions of people.

Redundancy is a very crucial feature that ensures the availability of information anytime and anywhere.

6.1 Types of Cloud Storage

There are three types of cloud storage:

1. _____. Public cloud storage delivers data storage infrastructure over the internet that is owned, maintained, and shared among multiple organizations by a third-party provider. Examples include Amazon S3, Google Cloud Storage, Microsoft Azure Blob Storage, Dropbox.
2. _____. Private cloud storage provides dedicated storage infrastructure used exclusively by a single organization, offering maximum security and control whether hosted on-premises or by a vendor. Examples include On-premises NAS/SAN arrays using Dell EMC PowerScale, VMware vSAN, AWS Outposts, or self-hosted Nextcloud.
3. _____. Hybrid cloud storage integrates both public and private cloud environments, allowing data and applications to be shared between them for optimal security, flexibility, and cost-efficiency. Examples include AWS Storage Gateway, Azure Stack integrated with Azure Blob Storage, NetApp Cloud Volumes ONTAP.

6.2 Infrastructure

Both public and private clouds use the same set of infrastructures, i.e., the same set of servers, storage, and virtualization techniques. The only difference is that these set of infrastructures are owned and managed in house in private cloud storage. The private cloud storage infrastructure is single tenant while the public cloud storages have multi-tenant infrastructures. However, enterprises can buy extra hardware and software components apart from these components. per the demand of the market.

The cloud storage infrastructure is elastic, scalable, and multi-tenant. It is not the same as it used to be in physical devices. Storage infrastructure can be accessed using the web service application programming interface.

6.3 Data Protection Process

_____. It means to follow the principals and policies that safeguard our data against corruption, loss, or damage. The data protection process involves techniques like backup, recovery and disaster recovery, and more.

The data protection process ensures the operation of various tasks like storage management, infrastructure security, and integrity. Many data protection techniques are developed to ensure a higher level of security but some risks or challenges always hamper the way to success. These challenges may be data loss or theft, replication of data, or insecure API.

Some measures that organizations can follow are as follows:

- Keeping a backup of data locally also, i.e., in their systems.
- Never storing sensitive data on cloud storage.
- Always using encrypted data to avoid data theft.

- Going for a cloud service that provides encryption and decryption facilities for your files and information. This allows only administrators and cloud service providers to access your files. No other person is allowed to access these files and information.
- Always using a strong password to avoid unauthorized access from anywhere outside.
- Never using a public computer and unsecured or open Wi-Fi hotspots to access your cloud storage. It may cause you data loss or theft from hackers or unauthorized persons. Also, take care of not disclosing your password on these connections.
- Going for an effective and well-versed antivirus for your system to protect your data from bugs and viruses.

6.4 Cloud Security

Cloud Security (or *Cloud Computing Security*). It refers to the set of policies and mechanisms that work together to provide a secure cloud environment. With a secure cloud-based system, users can work on and flourish their organizations and businesses.

The major risks that can hamper our cloud security are as follows:

- Risk of leakage or exposure of information to others.
- Risk of internal data of an organization being accessed by any unauthorized user or person who is not an employee of that particular organization. Any type of malicious attack that damages the cloud infrastructure results in data insecurity.
- Intrusion of a virus in the cloud storage.

6.5 Measures and Controls in Cloud Security

We can divide security controls into the following four categories:

1. Deterrent controls
2. Preventive controls
3. Detective controls
4. Corrective controls

6.5.1 Deterrent Controls

Deterrent Controls. Type of controls that give warnings to the attackers and make them aware that any act of data stealing or theft inside the cloud environment can be dangerous for them. These controls behave like a warning system and do not provide any type of security to the cloud environment or infrastructure. They are basically designed to discourage malicious activities by unauthorized people.

6.5.2 Preventive Controls

Preventive Controls. A preventive control actually includes a piece of code that performs the job of blocking inactive ports so as to prevent the hackers from entering the system. It enables strong authentication of users which protects the system from any kind unauthorized access. These types of controls are used for managing and preventing vulnerabilities in a cloud. By eliminating vulnerabilities, these controls help in strengthening the cloud environment.

6.5.3 Detective Controls

Detective Controls. Type of attacks that are used to identify the attack. They identify the threat and the events that occur on the online platform and provide their information to the concerned person. The users take the corrective action to resolve the problem caused by that particular suspicious threat or event.

In case of any suspicious activity, the detective control sends a signal to the preventive or corrective control in order to resolve the issue. Example of this type of attacks are intrusion detection and prevention.

6.5.4 Corrective Controls

Corrective Controls. The controls that mostly come into action during or after the attack has occurred. These controls limit the damage caused by the threat or attacker. The goal of these corrective controls is to reduce the consequences of the attack. Keeping a backup of the system is an example of a corrective control.

Apart from these controls, cloud providers can also use various techniques like encryption, cloud firewalls, backup plans, etc. Perfect security configurations for cloud providers and users and employee education are some other measures to ensure the safety of the cloud environment.

6.6 Encryption

Encryption. It is a process in which the data is encrypted in such a way that only the authorized users can access the data and information. Encryption uses a mathematical algorithm to transform plain text, i.e., a text, image, file or any other type of data into an encrypted form. This method enables users or companies to protect their data from getting stolen, damaged, or read by unauthorized or malicious users.